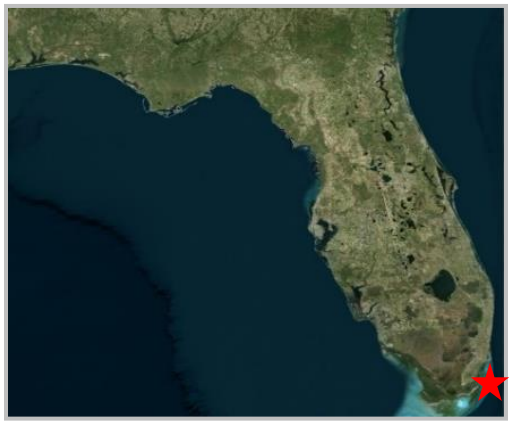
Oleta River State Park Estuarine Habitat Restoration Project

FY 20/21



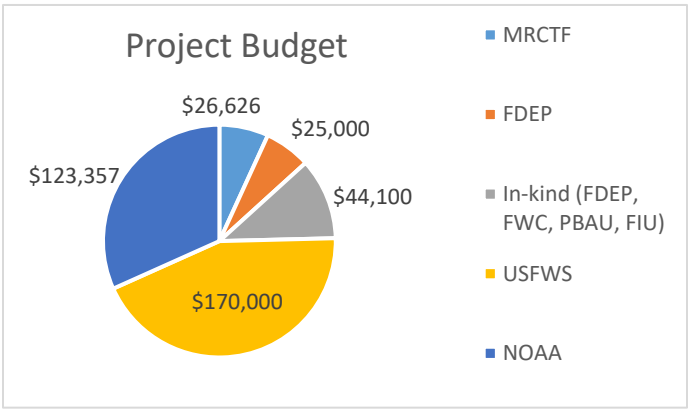


Figure 1: Oleta River State Park is in Biscayne Bay in Miami-Dade County in Southeast Florida.

Project Partners

- Florida Fish and Wildlife Conservation Commission (FWC)
- Florida Department of Environmental Protection (FDEP) Division of Recreation and Parks (DRP)
- US Fish and Wildlife Service (USFWS) State Wildlife Grants Program (SWG)
- National Oceanic and Atmospheric Administration (NOAA) Hurricane Irma Emergency Funds
- Florida International University (FIU)
- Palm Beach Atlantic University (PBAU)

Project Description

This project will restore two remnant mangrove systems, in Oleta River State Park, in Miami-Dade County, that are currently experiencing mangrove die-off. These wetlands were impounded within the landscape during the large-scale dredging and filling associated with coastal development in the 1950s and have been stressed due to failure of hydrologic connections (culverts) to tidal flow. This project will restore the two die-off areas, totaling 26.7 acres of mangrove habitat, by re-establishing tidal flow through the installation and maintenance of culverts and removal of hydrologic barriers.



Figure 2- Overview of the Oleta River Mangrove Restoration Project. The die-off areas to be restored are outlined in red.

Construction funding for this project was awarded through a grant from the USFWS SWG program administered by FWC. NOAA Hurricane Irma funds and FWC MRCTF provided funding for engineering, design and permitting of the project. The MRCT funding, and an additional $25,000, contributed by FDEP, will be used to match the SWG award. Specifically, in FY 20/21, MRCTF funds were used to fund underground utilities identification and mapping and project permitting. A portion of MRCTF funds also contributed to the purchase of 36" culvert pipes, purchased by FDEP and stored on site.

Match will also be provided by FDEP and FWC staff through in-kind services for project planning, monitoring, contract management, and grant management. Additionally, FIU and PBAU are providing in-kind services through water quality, prairie warbler, and Atlantic saltmarsh snake monitoring.

Benefits

Benefits to be realized from the final constructed project include:

- Directly restore and enhance, 26.7 acres for the benefit of fish and wildlife that are dependent on these species, specifically FWC Species of Greatest Conservation Need (SGCN) including Mangrove Saltmarsh Snake and Florida Prairie Warbler
- Improve water quality by restoring natural tidal regimes within the mangrove wetland
- Increase the wetlands capacity to function as a carbon sink by improving and increasing the extent of native vegetation, primarily mangroves.

MESS Strategic Plan Elements Supported

This proposal supports Objectives 1.1 and Strategic Actions 1.1.1 and 1.1.3 of Goal 1 of the MESS Strategic Plan.

Acreage restored by habitat type (footprint)

- 26.7 acres mangrove wetlands

Acreage affected by restoration activities (HUC 12): 15,409 acres



Figure 3: 296 linear feet of concrete pipe delivered to Oleta River State Park on May 6, 2021.



Figures 3: Mangrove saltmarsh snake captured, measured, and released on May 4, 2021 in the Oleta River State Park Mangrove Restoration project area.